



November 09, 2023

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: Semi-Annual PFAS
Pace Project No.: 50356219

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on October 12, 2023. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Minneapolis

If you have any questions concerning this report, please feel free to contact me.

Sincerely,

A handwritten signature in blue ink that reads "Brian Hall".

Brian Hall
brian.hall@pacelabs.com
(616)975-4500
Project Manager

Enclosures



REPORT OF LABORATORY ANALYSIS

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CERTIFICATIONS

Project: Semi-Annual PFAS

Pace Project No.: 50356219

Pace Analytical Services, LLC - Minneapolis MN

1700 Elm Street SE, Minneapolis, MN 55414

A2LA Certification #: 2926.01

Alabama Certification #: 40770

Alaska Contaminated Sites Certification #: 17-009

Alaska DW Certification #: MN00064

Arizona Certification #: AZ0014

Arkansas DW Certification #: MN00064

Arkansas WW Certification #: 88-0680

California Certification #: 2929

Colorado Certification #: MN00064

Connecticut Certification #: PH-0256

EPA Region 8 Tribal Water Systems+Wyoming DW
Certification #: via MN 027-053-137

Florida Certification #: E87605

Georgia Certification #: 959

GMP+ Certification #: GMP050884

Hawaii Certification #: MN00064

Idaho Certification #: MN00064

Illinois Certification #: 200011

Indiana Certification #: C-MN-01

Iowa Certification #: 368

Kansas Certification #: E-10167

Kentucky DW Certification #: 90062

Kentucky WW Certification #: 90062

Louisiana DEQ Certification #: AI-03086

Louisiana DW Certification #: MN00064

Maine Certification #: MN00064

Maryland Certification #: 322

Michigan Certification #: 9909

Minnesota Certification #: 027-053-137

Minnesota Dept of Ag Approval: via MN 027-053-137

Minnesota Petrofund Registration #: 1240

Mississippi Certification #: MN00064

Missouri Certification #: 10100

Montana Certification #: CERT0092

Nebraska Certification #: NE-OS-18-06

Nevada Certification #: MN00064

New Hampshire Certification #: 2081

New Jersey Certification #: MN002

New York Certification #: 11647

North Carolina DW Certification #: 27700

North Carolina WW Certification #: 530

North Dakota Certification (A2LA) #: R-036

North Dakota Certification (MN) #: R-036

Ohio DW Certification #: 41244

Ohio VAP Certification (1700) #: CL101

Oklahoma Certification #: 9507

Oregon Primary Certification #: MN300001

Oregon Secondary Certification #: MN200001

Pennsylvania Certification #: 68-00563

Puerto Rico Certification #: MN00064

South Carolina Certification #: 74003001

Tennessee Certification #: TN02818

Texas Certification #: T104704192

Utah Certification #: MN00064

Vermont Certification #: VT-027053137

Virginia Certification #: 460163

Washington Certification #: C486

West Virginia DEP Certification #: 382

West Virginia DW Certification #: 9952 C

Wisconsin Certification #: 999407970

Wyoming UST Certification #: via A2LA 2926.01

USDA Permit #: P330-19-00208

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SAMPLE SUMMARY

Project: Semi-Annual PFAS

Pace Project No.: 50356219

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50356219001	Effluent Duplicate	Water	10/11/23 13:37	10/12/23 10:15
50356219002	Effluent	Water	10/11/23 13:33	10/12/23 10:15
50356219003	Field Blank	Water	10/11/23 13:30	10/12/23 10:15

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SAMPLE ANALYTE COUNT

Project: Semi-Annual PFAS

Pace Project No.: 50356219

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
50356219001	Effluent Duplicate	ENV-SOP-MIN4-0178	NBH	48	PASI-M
50356219002	Effluent	ENV-SOP-MIN4-0178	NBH	48	PASI-M
50356219003	Field Blank	ENV-SOP-MIN4-0178	NBH	48	PASI-M

PASI-M = Pace Analytical Services - Minneapolis

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SUMMARY OF DETECTION

Project: Semi-Annual PFAS

Pace Project No.: 50356219

Lab Sample ID Method	Client Sample ID Parameters	Result	Units	Report Limit	Analyzed	Qualifiers
50356219001	Effluent Duplicate					
ENV-SOP-MIN4-0178	Perfluorobutanesulfonic acid	39.6	ng/L	3.9	11/07/23 19:21	
ENV-SOP-MIN4-0178	Perfluorohexanoic acid	14.2	ng/L	4.4	11/07/23 19:21	
50356219002	Effluent					
ENV-SOP-MIN4-0178	Perfluorobutanesulfonic acid	38.0	ng/L	3.8	11/07/23 19:28	
ENV-SOP-MIN4-0178	Perfluorohexanoic acid	14.0	ng/L	4.3	11/07/23 19:28	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Semi-Annual PFAS

Pace Project No.: 50356219

Sample: Effluent Duplicate		Lab ID: 50356219001		Collected: 10/11/23 13:37		Received: 10/12/23 10:15		Matrix: Water	
Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual	
PFAS ID NPW	Analytical Method: ENV-SOP-MIN4-0178 Preparation Method: ENV-SOP-MIN4-0178								
	Pace Analytical Services - Minneapolis								
11CI-PF3OUdS	<4.1	ng/L	4.1	1	11/03/23 10:35	11/07/23 19:21	763051-92-9		
4:2 FTS	<4.1	ng/L	4.1	1	11/03/23 10:35	11/07/23 19:21	757124-72-4		
6:2 FTS	<4.2	ng/L	4.2	1	11/03/23 10:35	11/07/23 19:21	27619-97-2		
8:2 FTS	<4.3	ng/L	4.3	1	11/03/23 10:35	11/07/23 19:21	39108-34-4		
9CI-PF3ONS	<4.1	ng/L	4.1	1	11/03/23 10:35	11/07/23 19:21	756426-58-1		
ADONA	<4.2	ng/L	4.2	1	11/03/23 10:35	11/07/23 19:21	919005-14-4		
HFPO-DA	<4.4	ng/L	4.4	1	11/03/23 10:35	11/07/23 19:21	13252-13-6		
NEtFOSAA	<4.4	ng/L	4.4	1	11/03/23 10:35	11/07/23 19:21	2991-50-6		
NMeFOSAA	<4.4	ng/L	4.4	1	11/03/23 10:35	11/07/23 19:21	2355-31-9		
Perfluorobutanesulfonic acid	39.6	ng/L	3.9	1	11/03/23 10:35	11/07/23 19:21	375-73-5		
Perfluorodecanoic acid	<4.4	ng/L	4.4	1	11/03/23 10:35	11/07/23 19:21	335-76-2		
Perfluorohexanoic acid	14.2	ng/L	4.4	1	11/03/23 10:35	11/07/23 19:21	307-24-4		
PFBA	<4.4	ng/L	4.4	1	11/03/23 10:35	11/07/23 19:21	375-22-4		
PFDS	<4.3	ng/L	4.3	1	11/03/23 10:35	11/07/23 19:21	335-77-3		
PFHpS	<4.2	ng/L	4.2	1	11/03/23 10:35	11/07/23 19:21	375-92-8		
PFNS	<4.2	ng/L	4.2	1	11/03/23 10:35	11/07/23 19:21	68259-12-1		
PFOSA	<4.4	ng/L	4.4	1	11/03/23 10:35	11/07/23 19:21	754-91-6		
PFPeA	<4.4	ng/L	4.4	1	11/03/23 10:35	11/07/23 19:21	2706-90-3		
PFPeS	<4.1	ng/L	4.1	1	11/03/23 10:35	11/07/23 19:21	2706-91-4		
Perfluorododecanoic acid	<4.4	ng/L	4.4	1	11/03/23 10:35	11/07/23 19:21	307-55-1		
Perfluoroheptanoic acid	<4.4	ng/L	4.4	1	11/03/23 10:35	11/07/23 19:21	375-85-9		
Perfluorohexanesulfonic acid	<4.0	ng/L	4.0	1	11/03/23 10:35	11/07/23 19:21	355-46-4		
Perfluorononanoic acid	<4.4	ng/L	4.4	1	11/03/23 10:35	11/07/23 19:21	375-95-1		
Perfluorooctanesulfonic acid	<4.1	ng/L	4.1	1	11/03/23 10:35	11/07/23 19:21	1763-23-1		
Perfluorooctanoic acid	<4.4	ng/L	4.4	1	11/03/23 10:35	11/07/23 19:21	335-67-1		
Perfluorotetradecanoic acid	<4.4	ng/L	4.4	1	11/03/23 10:35	11/07/23 19:21	376-06-7		
Perfluorotridecanoic acid	<4.4	ng/L	4.4	1	11/03/23 10:35	11/07/23 19:21	72629-94-8		
Perfluoroundecanoic acid	<4.4	ng/L	4.4	1	11/03/23 10:35	11/07/23 19:21	2058-94-8		
Surrogates									
13C26:2FTS (S)	270	%.	50-150	1	11/03/23 10:35	11/07/23 19:21		S0	
13C28:2FTS (S)	202	%.	50-150	1	11/03/23 10:35	11/07/23 19:21		S0	
13C24:2FTS (S)	199	%.	50-150	1	11/03/23 10:35	11/07/23 19:21		S0	
13C2-PFDoA (S)	80	%.	50-150	1	11/03/23 10:35	11/07/23 19:21			
13C2-PFTA (S)	75	%.	50-150	1	11/03/23 10:35	11/07/23 19:21			
13C3HFPO-DA (S)	46	%.	50-150	1	11/03/23 10:35	11/07/23 19:21		S0	
13C3-PFBS (S)	85	%.	50-150	1	11/03/23 10:35	11/07/23 19:21	375-73-5		
13C3-PFHxS (S)	88	%.	50-150	1	11/03/23 10:35	11/07/23 19:21	355-46-4		
13C4-PFBA (S)	74	%.	50-150	1	11/03/23 10:35	11/07/23 19:21	375-22-4		
13C4-PFHpA (S)	90	%.	50-150	1	11/03/23 10:35	11/07/23 19:21	375-85-9		
13C5-PFHxA (S)	90	%.	50-150	1	11/03/23 10:35	11/07/23 19:21	307-24-4		
13C5-PFPeA (S)	85	%.	50-150	1	11/03/23 10:35	11/07/23 19:21	2706-90-3		
13C6-PFDA (S)	90	%.	50-150	1	11/03/23 10:35	11/07/23 19:21	335-76-2		
13C7-PFUdA (S)	86	%.	50-150	1	11/03/23 10:35	11/07/23 19:21	2058-94-8		
13C8-PFOA (S)	94	%.	50-150	1	11/03/23 10:35	11/07/23 19:21	335-67-1		
13C8-PFOS (S)	83	%.	50-150	1	11/03/23 10:35	11/07/23 19:21	1763-23-1		
13C8-PFOSA (S)	72	%.	50-150	1	11/03/23 10:35	11/07/23 19:21	754-91-6		

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Semi-Annual PFAS

Pace Project No.: 50356219

Sample: Effluent Duplicate		Lab ID: 50356219001	Collected: 10/11/23 13:37	Received: 10/12/23 10:15	Matrix: Water			
Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
PFAS ID NPW		Analytical Method: ENV-SOP-MIN4-0178 Preparation Method: ENV-SOP-MIN4-0178 Pace Analytical Services - Minneapolis						
Surrogates								
13C9-PFNA (S)	94	%.	50-150	1	11/03/23 10:35	11/07/23 19:21	375-95-1	
d3-MeFOSAA (S)	89	%.	50-150	1	11/03/23 10:35	11/07/23 19:21	2355-31-9	
d5-EtFOSAA (S)	102	%.	50-150	1	11/03/23 10:35	11/07/23 19:21	2991-50-6	

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ANALYTICAL RESULTS

Project: Semi-Annual PFAS

Pace Project No.: 50356219

Sample: Effluent		Lab ID: 50356219002		Collected: 10/11/23 13:33		Received: 10/12/23 10:15		Matrix: Water	
Parameters		Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
PFAS ID NPW		Analytical Method: ENV-SOP-MIN4-0178 Preparation Method: ENV-SOP-MIN4-0178 Pace Analytical Services - Minneapolis							
11CI-PF3OUdS	<4.0	ng/L	4.0	1	11/03/23 10:35	11/07/23 19:28	763051-92-9		
4:2 FTS	<4.0	ng/L	4.0	1	11/03/23 10:35	11/07/23 19:28	757124-72-4		
6:2 FTS	<4.1	ng/L	4.1	1	11/03/23 10:35	11/07/23 19:28	27619-97-2		
8:2 FTS	<4.1	ng/L	4.1	1	11/03/23 10:35	11/07/23 19:28	39108-34-4		
9CI-PF3ONS	<4.0	ng/L	4.0	1	11/03/23 10:35	11/07/23 19:28	756426-58-1		
ADONA	<4.1	ng/L	4.1	1	11/03/23 10:35	11/07/23 19:28	919005-14-4		
HFPO-DA	<4.3	ng/L	4.3	1	11/03/23 10:35	11/07/23 19:28	13252-13-6		
NEtFOSAA	<4.3	ng/L	4.3	1	11/03/23 10:35	11/07/23 19:28	2991-50-6		
NMeFOSAA	<4.3	ng/L	4.3	1	11/03/23 10:35	11/07/23 19:28	2355-31-9		
Perfluorobutanesulfonic acid	38.0	ng/L	3.8	1	11/03/23 10:35	11/07/23 19:28	375-73-5		
Perfluorodecanoic acid	<4.3	ng/L	4.3	1	11/03/23 10:35	11/07/23 19:28	335-76-2		
Perfluorohexanoic acid	14.0	ng/L	4.3	1	11/03/23 10:35	11/07/23 19:28	307-24-4		
PFBA	<4.3	ng/L	4.3	1	11/03/23 10:35	11/07/23 19:28	375-22-4		
PFDS	<4.1	ng/L	4.1	1	11/03/23 10:35	11/07/23 19:28	335-77-3		
PFHpS	<4.1	ng/L	4.1	1	11/03/23 10:35	11/07/23 19:28	375-92-8		
PFNS	<4.1	ng/L	4.1	1	11/03/23 10:35	11/07/23 19:28	68259-12-1		
PFOSA	<4.3	ng/L	4.3	1	11/03/23 10:35	11/07/23 19:28	754-91-6		
PFPeA	<4.3	ng/L	4.3	1	11/03/23 10:35	11/07/23 19:28	2706-90-3		
PFPeS	<4.0	ng/L	4.0	1	11/03/23 10:35	11/07/23 19:28	2706-91-4		
Perfluorododecanoic acid	<4.3	ng/L	4.3	1	11/03/23 10:35	11/07/23 19:28	307-55-1		
Perfluoroheptanoic acid	<4.3	ng/L	4.3	1	11/03/23 10:35	11/07/23 19:28	375-85-9		
Perfluorohexanesulfonic acid	<3.9	ng/L	3.9	1	11/03/23 10:35	11/07/23 19:28	355-46-4		
Perfluorononanoic acid	<4.3	ng/L	4.3	1	11/03/23 10:35	11/07/23 19:28	375-95-1		
Perfluorooctanesulfonic acid	<4.0	ng/L	4.0	1	11/03/23 10:35	11/07/23 19:28	1763-23-1		
Perfluorooctanoic acid	<4.3	ng/L	4.3	1	11/03/23 10:35	11/07/23 19:28	335-67-1		
Perfluorotetradecanoic acid	<4.3	ng/L	4.3	1	11/03/23 10:35	11/07/23 19:28	376-06-7		
Perfluorotridecanoic acid	<4.3	ng/L	4.3	1	11/03/23 10:35	11/07/23 19:28	72629-94-8		
Perfluoroundecanoic acid	<4.3	ng/L	4.3	1	11/03/23 10:35	11/07/23 19:28	2058-94-8		
Surrogates									
13C26:2FTS (S)	255	%.	50-150	1	11/03/23 10:35	11/07/23 19:28			S0
13C28:2FTS (S)	185	%.	50-150	1	11/03/23 10:35	11/07/23 19:28			S0
13C24:2FTS (S)	173	%.	50-150	1	11/03/23 10:35	11/07/23 19:28			S0
13C2-PFDoA (S)	78	%.	50-150	1	11/03/23 10:35	11/07/23 19:28			
13C2-PFTA (S)	74	%.	50-150	1	11/03/23 10:35	11/07/23 19:28			
13C3HFPO-DA (S)	46	%.	50-150	1	11/03/23 10:35	11/07/23 19:28			S0
13C3-PFBS (S)	80	%.	50-150	1	11/03/23 10:35	11/07/23 19:28	375-73-5		
13C3-PFHxS (S)	83	%.	50-150	1	11/03/23 10:35	11/07/23 19:28	355-46-4		
13C4-PFBA (S)	70	%.	50-150	1	11/03/23 10:35	11/07/23 19:28	375-22-4		
13C4-PFHpA (S)	85	%.	50-150	1	11/03/23 10:35	11/07/23 19:28	375-85-9		
13C5-PFHxA (S)	84	%.	50-150	1	11/03/23 10:35	11/07/23 19:28	307-24-4		
13C5-PFPeA (S)	80	%.	50-150	1	11/03/23 10:35	11/07/23 19:28	2706-90-3		
13C6-PFDA (S)	84	%.	50-150	1	11/03/23 10:35	11/07/23 19:28	335-76-2		
13C7-PFUdA (S)	80	%.	50-150	1	11/03/23 10:35	11/07/23 19:28	2058-94-8		
13C8-PFOA (S)	84	%.	50-150	1	11/03/23 10:35	11/07/23 19:28	335-67-1		
13C8-PFOS (S)	81	%.	50-150	1	11/03/23 10:35	11/07/23 19:28	1763-23-1		
13C8-PFOSA (S)	65	%.	50-150	1	11/03/23 10:35	11/07/23 19:28	754-91-6		

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ANALYTICAL RESULTS

Project: Semi-Annual PFAS

Pace Project No.: 50356219

Sample: Effluent		Lab ID: 50356219002		Collected: 10/11/23 13:33		Received: 10/12/23 10:15		Matrix: Water	
Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual	
PFAS ID NPW									
Analytical Method: ENV-SOP-MIN4-0178 Preparation Method: ENV-SOP-MIN4-0178									
Pace Analytical Services - Minneapolis									
Surrogates									
13C9-PFNA (S)	86	%.	50-150	1	11/03/23 10:35	11/07/23 19:28	375-95-1		
d3-MeFOSAA (S)	83	%.	50-150	1	11/03/23 10:35	11/07/23 19:28	2355-31-9		
d5-EtFOSAA (S)	93	%.	50-150	1	11/03/23 10:35	11/07/23 19:28	2991-50-6		

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ANALYTICAL RESULTS

Project: Semi-Annual PFAS

Pace Project No.: 50356219

Sample: Field Blank		Lab ID: 50356219003		Collected: 10/11/23 13:30		Received: 10/12/23 10:15		Matrix: Water	
Parameters		Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
PFAS ID NPW		Analytical Method: ENV-SOP-MIN4-0178				Preparation Method: ENV-SOP-MIN4-0178			
		Pace Analytical Services - Minneapolis							
11CI-PF3OUdS	<1.9	ng/L	1.9	1	11/03/23 10:35	11/07/23 19:35	763051-92-9		
4:2 FTS	<1.9	ng/L	1.9	1	11/03/23 10:35	11/07/23 19:35	757124-72-4		
6:2 FTS	<1.9	ng/L	1.9	1	11/03/23 10:35	11/07/23 19:35	27619-97-2		
8:2 FTS	<1.9	ng/L	1.9	1	11/03/23 10:35	11/07/23 19:35	39108-34-4		
9CI-PF3ONS	<1.9	ng/L	1.9	1	11/03/23 10:35	11/07/23 19:35	756426-58-1		
ADONA	<1.9	ng/L	1.9	1	11/03/23 10:35	11/07/23 19:35	919005-14-4		
HFPO-DA	<2.0	ng/L	2.0	1	11/03/23 10:35	11/07/23 19:35	13252-13-6		
NEtFOSAA	<2.0	ng/L	2.0	1	11/03/23 10:35	11/07/23 19:35	2991-50-6		
NMeFOSAA	<2.0	ng/L	2.0	1	11/03/23 10:35	11/07/23 19:35	2355-31-9		
Perfluorobutanesulfonic acid	<1.8	ng/L	1.8	1	11/03/23 10:35	11/07/23 19:35	375-73-5		
Perfluorodecanoic acid	<2.0	ng/L	2.0	1	11/03/23 10:35	11/07/23 19:35	335-76-2		
Perfluorohexanoic acid	<2.0	ng/L	2.0	1	11/03/23 10:35	11/07/23 19:35	307-24-4		
PFBA	<2.0	ng/L	2.0	1	11/03/23 10:35	11/07/23 19:35	375-22-4		
PFDS	<1.9	ng/L	1.9	1	11/03/23 10:35	11/07/23 19:35	335-77-3		
PFHpS	<1.9	ng/L	1.9	1	11/03/23 10:35	11/07/23 19:35	375-92-8		
PFNS	<1.9	ng/L	1.9	1	11/03/23 10:35	11/07/23 19:35	68259-12-1		
PFOSA	<2.0	ng/L	2.0	1	11/03/23 10:35	11/07/23 19:35	754-91-6		
PFPeA	<2.0	ng/L	2.0	1	11/03/23 10:35	11/07/23 19:35	2706-90-3		
PFPeS	<1.9	ng/L	1.9	1	11/03/23 10:35	11/07/23 19:35	2706-91-4		
Perfluorododecanoic acid	<2.0	ng/L	2.0	1	11/03/23 10:35	11/07/23 19:35	307-55-1		
Perfluoroheptanoic acid	<2.0	ng/L	2.0	1	11/03/23 10:35	11/07/23 19:35	375-85-9		
Perfluorohexanesulfonic acid	<1.8	ng/L	1.8	1	11/03/23 10:35	11/07/23 19:35	355-46-4		
Perfluorononanoic acid	<2.0	ng/L	2.0	1	11/03/23 10:35	11/07/23 19:35	375-95-1		
Perfluorooctanesulfonic acid	<1.9	ng/L	1.9	1	11/03/23 10:35	11/07/23 19:35	1763-23-1		
Perfluorooctanoic acid	<2.0	ng/L	2.0	1	11/03/23 10:35	11/07/23 19:35	335-67-1		
Perfluorotetradecanoic acid	<2.0	ng/L	2.0	1	11/03/23 10:35	11/07/23 19:35	376-06-7		
Perfluorotridecanoic acid	<2.0	ng/L	2.0	1	11/03/23 10:35	11/07/23 19:35	72629-94-8		
Perfluoroundecanoic acid	<2.0	ng/L	2.0	1	11/03/23 10:35	11/07/23 19:35	2058-94-8		
Surrogates									
13C26:2FTS (S)	62	%.	50-150	1	11/03/23 10:35	11/07/23 19:35			
13C28:2FTS (S)	53	%.	50-150	1	11/03/23 10:35	11/07/23 19:35			
13C24:2FTS (S)	52	%.	50-150	1	11/03/23 10:35	11/07/23 19:35			
13C2-PFDoA (S)	49	%.	50-150	1	11/03/23 10:35	11/07/23 19:35	S0		
13C2-PFTA (S)	48	%.	50-150	1	11/03/23 10:35	11/07/23 19:35	S0		
13C3HFPO-DA (S)	80	%.	50-150	1	11/03/23 10:35	11/07/23 19:35			
13C3-PFBS (S)	98	%.	50-150	1	11/03/23 10:35	11/07/23 19:35	375-73-5		
13C3-PFHxS (S)	91	%.	50-150	1	11/03/23 10:35	11/07/23 19:35	355-46-4		
13C4-PFBA (S)	99	%.	50-150	1	11/03/23 10:35	11/07/23 19:35	375-22-4		
13C4-PFHpA (S)	97	%.	50-150	1	11/03/23 10:35	11/07/23 19:35	375-85-9		
13C5-PFHxA (S)	98	%.	50-150	1	11/03/23 10:35	11/07/23 19:35	307-24-4		
13C5-PFPeA (S)	99	%.	50-150	1	11/03/23 10:35	11/07/23 19:35	2706-90-3		
13C6-PFDA (S)	58	%.	50-150	1	11/03/23 10:35	11/07/23 19:35	335-76-2		
13C7-PFUdA (S)	52	%.	50-150	1	11/03/23 10:35	11/07/23 19:35	2058-94-8		
13C8-PFOA (S)	85	%.	50-150	1	11/03/23 10:35	11/07/23 19:35	335-67-1		
13C8-PFOS (S)	57	%.	50-150	1	11/03/23 10:35	11/07/23 19:35	1763-23-1		
13C8-PFOSA (S)	63	%.	50-150	1	11/03/23 10:35	11/07/23 19:35	754-91-6		

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Semi-Annual PFAS

Pace Project No.: 50356219

Sample: Field Blank		Lab ID: 50356219003		Collected: 10/11/23 13:30		Received: 10/12/23 10:15		Matrix: Water	
Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual	
PFAS ID NPW		Analytical Method: ENV-SOP-MIN4-0178 Preparation Method: ENV-SOP-MIN4-0178 Pace Analytical Services - Minneapolis							
Surrogates									
13C9-PFNA (S)	69	%.	50-150	1	11/03/23 10:35	11/07/23 19:35	375-95-1		
d3-MeFOSAA (S)	57	%.	50-150	1	11/03/23 10:35	11/07/23 19:35	2355-31-9		
d5-EtFOSAA (S)	58	%.	50-150	1	11/03/23 10:35	11/07/23 19:35	2991-50-6		

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QUALITY CONTROL DATA

Project: Semi-Annual PFAS

Pace Project No.: 50356219

QC Batch: 915320 Analysis Method: ENV-SOP-MIN4-0178
QC Batch Method: ENV-SOP-MIN4-0178 Analysis Description: PFAS Compounds, Water
Laboratory: Pace Analytical Services - Minneapolis
Associated Lab Samples: 50356219001, 50356219002, 50356219003

METHOD BLANK: 4816491 Matrix: Water

Associated Lab Samples: 50356219001, 50356219002, 50356219003

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
11CI-PF3OUdS	ng/L	<1.8	1.8	11/07/23 18:59	
4:2 FTS	ng/L	<1.8	1.8	11/07/23 18:59	
6:2 FTS	ng/L	<1.8	1.8	11/07/23 18:59	
8:2 FTS	ng/L	<1.9	1.9	11/07/23 18:59	
9CI-PF3ONS	ng/L	<1.8	1.8	11/07/23 18:59	
ADONA	ng/L	<1.8	1.8	11/07/23 18:59	
HFPO-DA	ng/L	<1.9	1.9	11/07/23 18:59	
NEtFOSAA	ng/L	<1.9	1.9	11/07/23 18:59	
NMeFOSAA	ng/L	<1.9	1.9	11/07/23 18:59	
Perfluorobutanesulfonic acid	ng/L	<1.7	1.7	11/07/23 18:59	
Perfluorodecanoic acid	ng/L	<1.9	1.9	11/07/23 18:59	
Perfluorododecanoic acid	ng/L	<1.9	1.9	11/07/23 18:59	
Perfluoroheptanoic acid	ng/L	<1.9	1.9	11/07/23 18:59	
Perfluorohexanesulfonic acid	ng/L	<1.8	1.8	11/07/23 18:59	
Perfluorohexanoic acid	ng/L	<1.9	1.9	11/07/23 18:59	
Perfluorononanoic acid	ng/L	<1.9	1.9	11/07/23 18:59	
Perfluorooctanesulfonic acid	ng/L	<1.8	1.8	11/07/23 18:59	
Perfluorooctanoic acid	ng/L	<1.9	1.9	11/07/23 18:59	
Perfluorotetradecanoic acid	ng/L	<1.9	1.9	11/07/23 18:59	
Perfluorotridecanoic acid	ng/L	<1.9	1.9	11/07/23 18:59	
Perfluoroundecanoic acid	ng/L	<1.9	1.9	11/07/23 18:59	
PFBA	ng/L	<1.9	1.9	11/07/23 18:59	
PFDS	ng/L	<1.9	1.9	11/07/23 18:59	
PFHpS	ng/L	<1.8	1.8	11/07/23 18:59	
PFNS	ng/L	<1.8	1.8	11/07/23 18:59	
PFOSA	ng/L	<1.9	1.9	11/07/23 18:59	
PFPeA	ng/L	<1.9	1.9	11/07/23 18:59	
PFPeS	ng/L	<1.8	1.8	11/07/23 18:59	
13C2-PFDoA (S)	%	68	50-150	11/07/23 18:59	
13C2-PFTA (S)	%	64	50-150	11/07/23 18:59	
13C24:2FTS (S)	%	87	50-150	11/07/23 18:59	
13C26:2FTS (S)	%	88	50-150	11/07/23 18:59	
13C28:2FTS (S)	%	74	50-150	11/07/23 18:59	
13C2PFHxDA (S)	%	38	50-150	11/07/23 18:59	S0
13C3-PFBS (S)	%	108	50-150	11/07/23 18:59	
13C3-PFHxS (S)	%	106	50-150	11/07/23 18:59	
13C3HFPO-DA (S)	%	88	50-150	11/07/23 18:59	
13C4-PFBA (S)	%	110	50-150	11/07/23 18:59	
13C4-PFHpA (S)	%	106	50-150	11/07/23 18:59	
13C5-PFHxA (S)	%	107	50-150	11/07/23 18:59	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Semi-Annual PFAS

Pace Project No.: 50356219

METHOD BLANK: 4816491

Matrix: Water

Associated Lab Samples: 50356219001, 50356219002, 50356219003

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
13C5-PFPeA (S)	%.	108	50-150	11/07/23 18:59	
13C6-PFDA (S)	%.	75	50-150	11/07/23 18:59	
13C7-PFUDA (S)	%.	70	50-150	11/07/23 18:59	
13C8-PFOA (S)	%.	101	50-150	11/07/23 18:59	
13C8-PFOS (S)	%.	74	50-150	11/07/23 18:59	
13C8-PFOSA (S)	%.	84	50-150	11/07/23 18:59	
13C9-PFNA (S)	%.	87	50-150	11/07/23 18:59	
d3-MeFOSAA (S)	%.	74	50-150	11/07/23 18:59	
d3-NMeFOSA (S)	%.	58	10-150	11/07/23 18:59	
d5-EtFOSAA (S)	%.	80	50-150	11/07/23 18:59	
d5-NEtFOSA (S)	%.	53	10-150	11/07/23 18:59	
d7-NMeFOSE (S)	%.	67	10-150	11/07/23 18:59	
d9-NEtFOSE (S)	%.	65	10-150	11/07/23 18:59	

LABORATORY CONTROL SAMPLE & LCSD: 4816492

4819251

Parameter	Units	Spike Conc.	LCS Result	LCSD Result	LCS % Rec	LCSD % Rec	% Rec Limits	RPD	Max RPD	Qualifiers
11CI-PF3OUdS	ng/L	3.8	3.2	2.8	84	76	70-140	15	30	
4:2 FTS	ng/L	3.8	3.7	3.6	96	101	63-143	0	30	
6:2 FTS	ng/L	3.8	3.5	3.4	90	92	64-140	3	30	
8:2 FTS	ng/L	3.9	4.4	3.3	114	88	67-138	30	30	
9CI-PF3ONS	ng/L	3.8	3.4	2.9	90	81	70-140	15	30	
ADONA	ng/L	3.8	3.9	3.8	102	103	70-140	3	30	
HFPO-DA	ng/L	4	5.1	4.8	125	124	70-140	6	30	
NEtFOSAA	ng/L	4	3.7	3.5	91	90	61-135	5	30	
NMeFOSAA	ng/L	4	3.6	3.4	90	87	65-136	7	30	
Perfluorobutanesulfonic acid	ng/L	3.6	3.5	3.3	97	97	72-130	4	30	
Perfluorodecanoic acid	ng/L	4	3.7	3.5	92	89	71-129	8	30	
Perfluorododecanoic acid	ng/L	4	3.6	3.8	89	97	72-134	5	30	
Perfluoroheptanoic acid	ng/L	4	3.7	3.6	91	93	72-130	2	30	
Perfluorohexanesulfonic acid	ng/L	3.7	3.5	3.4	94	95	68-131	4	30	
Perfluorohexanoic acid	ng/L	4	4.0	4.0	100	103	72-129	1	30	
Perfluorononanoic acid	ng/L	4	3.6	3.7	89	96	69-130	4	30	
Perfluorooctanesulfonic acid	ng/L	3.8	3.4	3.4	89	96	65-140	2	30	
Perfluorooctanoic acid	ng/L	4	4.0	3.8	98	99	71-133	3	30	
Perfluorotetradecanoic acid	ng/L	4	3.7	3.9	91	101	71-132	6	30	
Perfluorotridecanoic acid	ng/L	4	3.6	3.6	88	93	65-144	1	30	
Perfluoroundecanoic acid	ng/L	4	3.8	3.3	94	86	69-133	14	30	
PFBA	ng/L	4	3.9	3.8	96	98	73-129	3	30	
PFDS	ng/L	3.9	3.4	2.4	87	63	53-142	36	30	R1
PFHpS	ng/L	3.9	4.5	4.5	118	122	69-134	0	30	
PFNS	ng/L	3.9	3.7	2.9	96	78	69-127	25	30	
PFOSA	ng/L	4	3.9	3.9	97	100	67-137	1	30	
PFPeA	ng/L	4	3.9	3.8	97	97	72-129	4	30	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Semi-Annual PFAS

Pace Project No.: 50356219

LABORATORY CONTROL SAMPLE & LCSD:		4816492	4819251							
Parameter	Units	Spike Conc.	LCS Result	LCSD Result	LCS % Rec	LCSD % Rec	% Rec Limits	RPD	Max RPD	Qualifiers
PFPeS	ng/L	3.8	3.8	3.4	99	95	71-127	9	30	
13C2-PFDaA (S)	%.				80	71	50-150			
13C2-PFTA (S)	%.				70	65	50-150			
13C24:2FTS (S)	%.				87	86	50-150			
13C26:2FTS (S)	%.				85	86	50-150			
13C28:2FTS (S)	%.				60	74	50-150			
13C2PFHxDA (S)	%.				47	35	50-150			S0
13C3-PFBS (S)	%.				105	104	50-150			
13C3-PFHxS (S)	%.				102	102	50-150			
13C3HFPO-DA (S)	%.				83	81	50-150			
13C4-PFBA (S)	%.				106	103	50-150			
13C4-PFHpA (S)	%.				105	104	50-150			
13C5-PFHxA (S)	%.				105	103	50-150			
13C5-PFPeA (S)	%.				105	104	50-150			
13C6-PFDA (S)	%.				84	78	50-150			
13C7-PFUdA (S)	%.				77	73	50-150			
13C8-PFOA (S)	%.				98	96	50-150			
13C8-PFOS (S)	%.				81	86	50-150			
13C8-PFOSA (S)	%.				91	87	50-150			
13C9-PFNA (S)	%.				93	92	50-150			
d3-MeFOSAA (S)	%.				82	79	50-150			
d3-NMeFOSA (S)	%.				54	56	10-150			
d5-EtFOSAA (S)	%.				87	80	50-150			
d5-NEtFOSA (S)	%.				50	47	10-150			
d7-NMeFOSE (S)	%.				68	61	10-150			
d9-NEtFOSE (S)	%.				68	62	10-150			

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QUALIFIERS

Project: Semi-Annual PFAS
Pace Project No.: 50356219

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.
ND - Not Detected at or above adjusted reporting limit.
TNTC - Too Numerous To Count
J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.
MDL - Adjusted Method Detection Limit.
PQL - Practical Quantitation Limit.
RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.
S - Surrogate
1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.
Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.
LCS(D) - Laboratory Control Sample (Duplicate)
MS(D) - Matrix Spike (Duplicate)
DUP - Sample Duplicate
RPD - Relative Percent Difference
NC - Not Calculable.
SG - Silica Gel - Clean-Up
U - Indicates the compound was analyzed for, but not detected.
N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.
Reported results are not rounded until the final step prior to reporting. Therefore, calculated parameters that are typically reported as "Total" may vary slightly from the sum of the reported component parameters.
Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.
TNI - The NELAC Institute.

ANALYTE QUALIFIERS

R1 RPD value was outside control limits.
S0 Surrogate recovery outside laboratory control limits.

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: Semi-Annual PFAS
Pace Project No.: 50356219

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50356219001	Effluent Duplicate	ENV-SOP-MIN4-0178	915320	ENV-SOP-MIN4-0178	917219
50356219002	Effluent	ENV-SOP-MIN4-0178	915320	ENV-SOP-MIN4-0178	917219
50356219003	Field Blank	ENV-SOP-MIN4-0178	915320	ENV-SOP-MIN4-0178	917219

REPORT OF LABORATORY ANALYSIS



Sample Conditions Upon Receipt Form (SCUR)

Date/Time: 10-12-23	Evaluated By: WDC	WO#: 50356219	
Client: City of Cadillac	PM: BJH	PM: BJH	Due Date: 11/10/23
Lab Notified of Rush or Short Holds: YES NO NA		CLIENT: GR-Cadillac	
Project Received Via: FedEx UPS Client Pace Courier Other: _____			Comments:
Custody Seal Present and Intact:	YES	NO	NA
Received Sample Information Form (SIF): Drinking Waters Only	YES	NO	NA
Short Hold Present (≤ 48 Hours):	YES	NO	
Sample Received in Hold:	YES	NO	
Custody Signature Present:	YES	NO	
Collector Signature Present:	YES	NO	
Sample Collected Today and On Ice:	YES	NO	N/A
IR Gun #: 350 351 Therm #: 282 283	Temp. should be 0°C - 6°C (Initial/Corrected)		
Ice Type: WET Bagged / WET Loose BLUE NONE	1. Cooler Temp. Upon Receipt: 4.8/4.5 °C		
Ice Location: TOP BOTTOM MIDDLE DISPERSED	2. Cooler Temp. Upon Receipt: _____ °C		
Temp Blank Received:	YES	NO	
Sample Label Matches COC (ID/Date/Time):	YES	NO	
Container Intact:	YES	NO	
Correct Container:	YES	NO	
Sufficient Volume:	YES	NO	
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation. Denote with red dot on container lid if unacceptable. pH Strip Lot #: _____ Exceptions are VOA, coliform, LLHg, O&G/TPH, or any container with a septum cap or preserved with HCl	YES	NO	N/A
Residual Chlorine Absent: Cl ₂ Strip Lot #: _____ Applies to SVOC 625, PCB/Pest. 608, Total/Amenable/Available/Free Cyanide	YES	NO	N/A
VOA Headspace Acceptable (<6mm): Denote with silver x on rim of container cap if unacceptable.	YES	NO	N/A
Trip Blank Received: HCl MeOH Other: _____	YES	NO	ON HOLD
Comments:	3. Cooler Temp. Upon Receipt: _____ °C		
	4. Cooler Temp. Upon Receipt: _____ °C		
	Non-Conformance Form Required: YES NO		